

Taylor Approximation Visual

A degree-by-degree visual showing how local derivative data builds a Taylor approximation.

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Taylor approximation sequence

1. Choose a center
2. Compute derivatives or known series
3. Build the degree-n polynomial
4. State the interval
5. Bound the remainder

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Taylor construction

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$$
$$R_n(x) = f(x) - T_n(x)$$